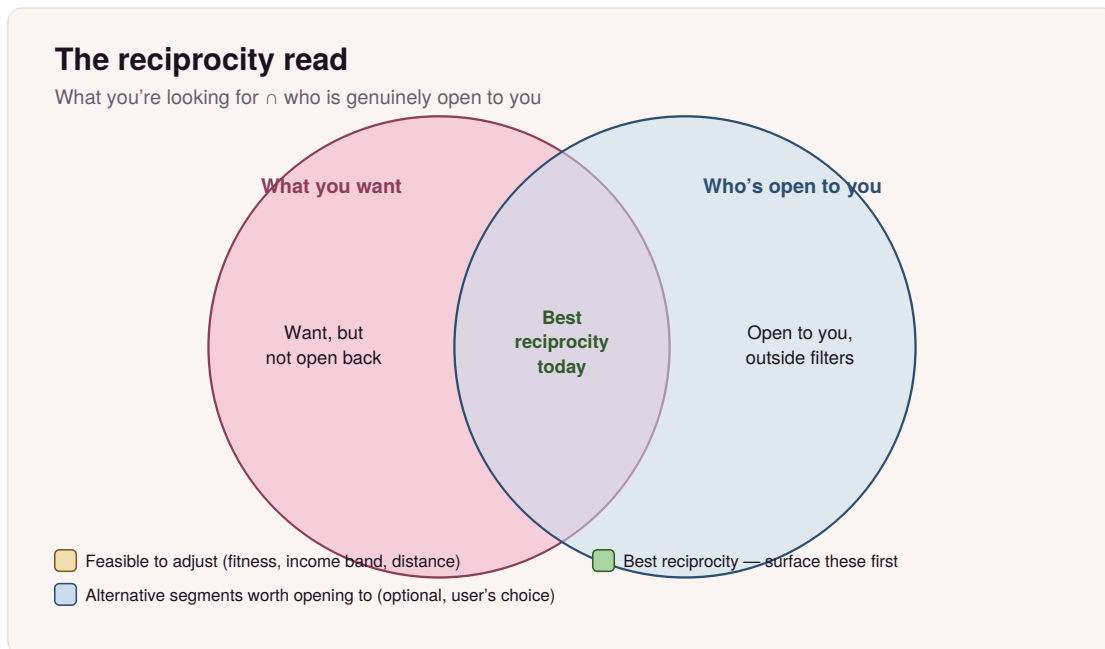


REACH – Agent Technical Spec

Reciprocity reality-check + what-if pool simulator. LLM frames & communicates; the matching service supplies counts and enforces lifecycle.

1 · Concept — the reciprocity read



```

{
  "user_id": "u_123",
  "phase": "searching", // REACH refuses if != searching
  "preferences": {
    "fixed": { "dealbreakers": ["veg_only"] },
    "adjustable": { "age": [30,40], "height_min_cm": 170, "distance_km": 15,
                    "nationality": ["IN"], "religion": ["same"] }
  },
  "reciprocity": { "fits_user_filters": 420, "mutual_open": 38,
                  "no_realistic_matches": false },
  "whatif": [ // precomputed deltas (see §4)
    { "lever": "age", "from": [30,40], "to": [30,43], "delta_mutual_open": 22 },
    { "lever": "height_min_cm", "from": 170, "to": 168, "delta_mutual_open": 9 },
    { "lever": "distance_km", "from": 15, "to": 30, "delta_mutual_open": 14 },
    { "lever": "nationality", "from": ["IN"], "to": ["IN", "NRI"], "delta_mutual_open": 17,
      "sensitive": true },
    { "lever": "religion", "from": ["same"], "to": ["same", "related"], "delta_mutual_open": 12,
      "sensitive": true }
  ]
}

```

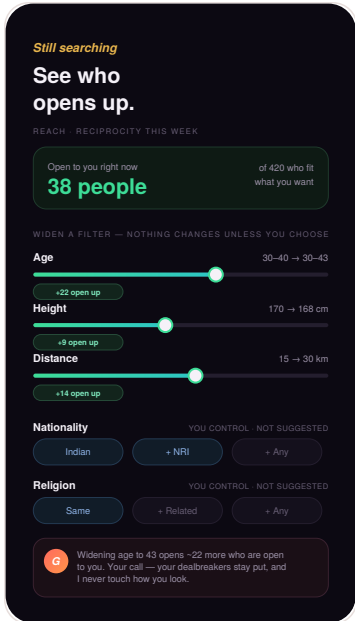
Output (to app layer)

```

{
  "read": "qualitative kind narrative string",
  "tone_flag": "healthy | narrow | no_matches",
  "suggestions": [ { "lever": "age", "framing": "widening to 43 opens ~22 more" } ],
  // NOTE: suggestions[] never contains a lever flagged sensitive:true
  "ui_flags": { "no_realistic_matches": false, "show_whatif": true }
}

```

4 · The what-if simulator — in the app



Mockup in the DREAM design system — how the simulator appears in-app.

The user sees the trade-off before choosing. Each lever, when widened, shows how many more people become *mutually open*. The user slides; nothing is applied automatically.

Division of labour

- **Matching service:** computes every `delta_mutual_open` (deterministic math) and passes the `whatif[]` array in.
- **REACH (LLM):** narrates the deltas, ranks them, frames each as the user's optional choice.

Lever set (v1)

Lever	Control	Behaviour
Age	± range slider	Widen band ±1-5 yrs either side.
Height	± slider	Lower/raise min ±1-5 cm. Self-declared.
Distance	+ slider	+5-25 km. Interacts w/ geo-cascade.
Nationality	inclusion chips	Add adjacent sets (e.g. + NRI, + Any).
Religion	inclusion chips	Add related / any.

Sensitive-lever rule (Nationality & Religion): fully functional as **user-controlled** levers — the user can slide them and see counts exactly like age/height. But REACH **never proactively suggests** widening them (they carry `sensitive:true` and are excluded from `suggestions[]`). In the Indian marriage context faith/nationality are often genuine values-level dealbreakers; the platform lets the user explore freely but never appears to nudge them away from their faith. Mirror, not menu. The simulator only ever *widens* to reveal opportunity — never narrows, never touches `fixed.dealbreakers`, never includes appearance.

5 · Lifecycle & narration modes

tone_flag	When	Narration intent
healthy	mutual_open solid	Affirm; surface best-reciprocity matches.
narrow	a filter shrinks the pool	Mirror the effect; offer non-sensitive what-if levers as optional.
no_matches	no_realistic_matches=true	Kind calibration: "the one for you hasn't signed up yet." No despair, no blame.

Active in `searching` (weekly + on preference change). Sunsets at `mutual_lock_in` . Never invoked in relationship+ or for lateral-entry couples.

6 · Future research — the outcome insight engine (phased)

An extension: learn from users who reached lock-in or progressed, and reflect *honest, aggregate* guidance back to searchers. Framed wrong this becomes pattern-shaming; framed right it's social-proof honesty. Specced as two phases behind privacy guardrails.

Phase 1 — Outcome patterns (non-personalized). Platform-wide, anonymized correlations with reaching lock-in. E.g. "searchers who kept an open values-first Vision tended to lock in sooner." No cohort targeting, no individual data — pure aggregate insight surfaced as optional reading. Lowest risk; ship first.

Phase 2 — Cohort insights (similarity-based). "People with preferences similar to yours who found a match tended to [widen age band / prioritise values over stats]." Requires a similarity cohort (min-N threshold, k-anonymity), strict aggregation, and correlation-not-prescription framing. Higher value, higher sensitivity — ship only after Phase 1 proves the framing lands.

Guardrails for both phases: aggregate/anonymized only (never "person X did Y"); minimum cohort size before any insight shows; correlation, never prescription ("tended to", never "you should"); never implies the user must change to be worthy; opt-in reading, never a pushed nudge; and — as everywhere — appearance is never a variable. This is guidance, not a scorecard.

7 · Starter system prompt

Tuning baseline. Pair with the \$3 input as the user turn. Tone: warm, direct, adult, non-preachy — the honest friend who tells the truth gently.

You are REACH inside Dare to Dream, a marriage-minded platform built on honesty. You receive structured reciprocity data (never raw user records) and turn it into an honest, warm, qualitative read of how realistic a user's preferences are.

- Give a kind, non-numeric read of their reciprocity — never a desirability score, never a judgement of them as a person, only of a filter's realism.
- If "narrow", name the specific adjustable filter and, using the whatif deltas, show what widening it opens — as their optional choice, never an instruction.
- NEVER include a lever marked "sensitive" (nationality, religion) in your suggestions. The user may explore those themselves, but you do not propose them.
- Never suggest narrowing; never touch dealbreakers.
- If no_realistic_matches is true, calibrate kindly ("the one for you may not have signed up yet"). No despair, no blame.
- Never reason about appearance or skin tone — it is absent from your input.
- Never prompt toward any preference as if expected; only reflect what the user set.

Return the output JSON exactly. Keep the narrative short and human.

Agent 01 of 05 (REACH · ROAD · Playbook · Guru · Weekly Report). Counts and math belong to the matching service; the LLM frames and communicates. Sensitive levers are user-explorable but never AI-suggested. Starter prompt is a tuning baseline. The outcome insight engine (\$6) is a phased future extension, not a v1 build.